1.32 Differentiation: Tangents and Normals

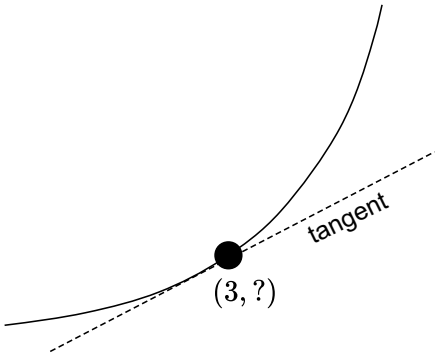
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Finding equations of tangents

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Find the equation of the **tangent** to the curve $y = x^2$ when $x = 3$.

We want to use $y - y_1 = m\,(x - x_1)$ for the tangent (as it is a straight line!).
Therefore we need:
(a) A point (x_1, y_1)
(b) The gradient m.



Gradient function:

Gradient when $x = 3$:

y-value when $x = 3$:

So equation of tangent:

$$\mathbf{y-9=6\,(x-3)}$$

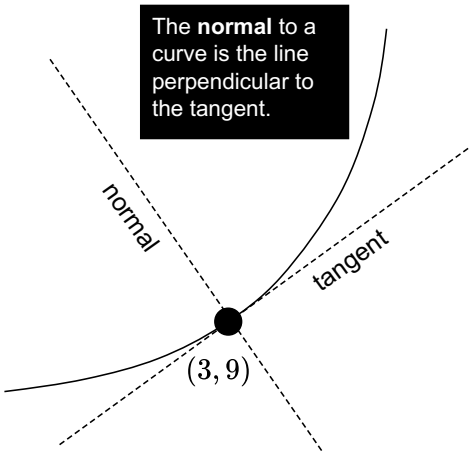
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Finding equations of normals

29

Find the equation of the **normal** to the curve $y = x^2$ when $x = 3$.

The **normal** to a curve is the line perpendicular to the tangent.



Equation of tangent (from earlier):

$$\mathbf{y-9=6(x-3)}$$

Therefore equation of normal:

$$\mathbf{y-9=}\ \tfrac{1}{\mathbf{6}}\mathbf{(x-3)}$$

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Test Your Understanding

30

Find the equation of the **normal** to the curve $y = x + 3\sqrt{x}$ when $x = 9$.

When $x = 9$, $y =$

$x + 3x^{\frac{1}{2}}$

$1 + \frac{3}{2}x^{-\frac{1}{2}}$

Tip: I like to use m_T and m_N to make clear to the examiner (and myself) what gradient I've found.

$= -\frac{2}{3}(x -$

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Exercise 12F – Pearson Pure Y1 page 269-270

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3 Find the coordinates of the point where the tangent to the curve $y = x^2 + 1$ at the point $(2, 5)$ meets the normal to the same curve at the point $(1, 2)$.

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Exercise 12F – Pearson Pure Y1 page 269-270

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4 Find the equations of the normals to the curve $y = x + x^3$ at the points $(0, 0)$ and $(1, 2)$, and find the coordinates of the point where these normals meet.

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Exercise 12F – Pearson Pure Y1 page 269-270

33

5 For $\mathrm{f}(x) = 12 - 4x + 2x^2$, find the equations of the tangent and the normal at the point where $x = -1$ on the curve with equation $y = \mathrm{f}(x)$.

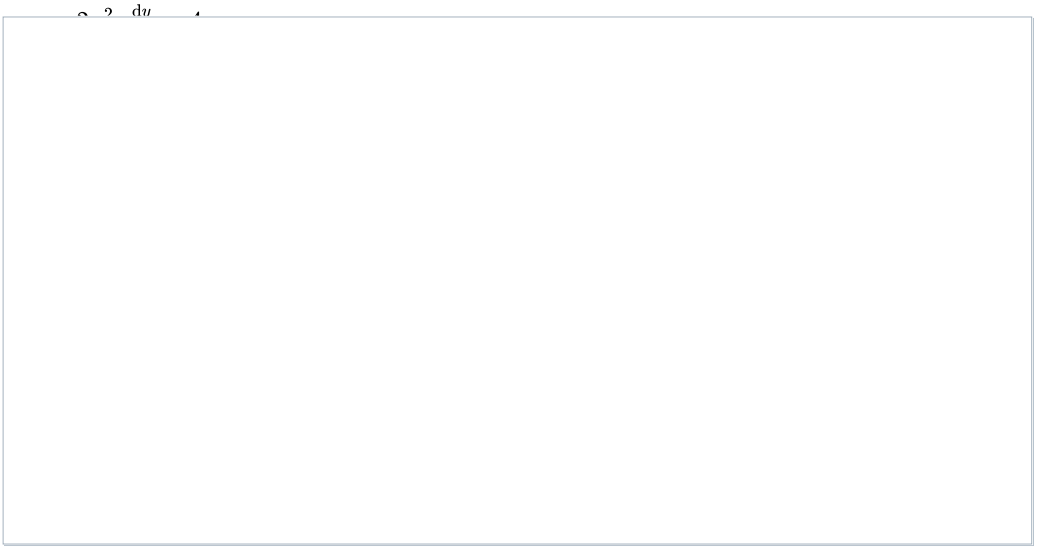
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Exercise 12F – Pearson Pure Y1 page 269-270

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6 The point P with x-coordinate $\frac{1}{2}$ lies on the curve with equation $y = 2x^2$. The normal to the curve at P intersects the curve at points P and Q. Find the coordinates of Q. (6 marks)



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Extension

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The 'difference of two cubes':
$a^3 - b^3 = (a - b)\left(a^2 + ab + b^2\right)$
will help for both of these.

[STEP I 2005 Q2]
The point P has coordinates $\left(p^2, 2p\right)$ and the point Q has coordinates $\left(q^2, 2q\right)$, where p and q are non-zero and $p \neq q$. The curve C is given by $y^2 = 4x$. The point R is the intersection of the tangent to C at P and the tangent to C at Q. Show that R has coordinates $(pq, p + q)$.

The point S is the intersection of the normal to C at P and the normal to C at Q. If p and q are such that $(1, 0)$ lies on the line PQ, show that S has coordinates $\left(p^2 + q^2 + 1, p + q\right)$, and that the quadrilateral $PSQR$ is a rectangle.

Solutions on next slide.

[STEP I 2012 Q4]
The curve C has equation $xy = \frac{1}{2}$. The tangents to C at the distinct points $P\left(p, \frac{1}{2p}\right)$ and $Q\left(q, \frac{1}{2q}\right)$, where p and q are positive, intersect at T and the normal to C at these points intersect at N. Show that T is the point $\left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$
In the case $pq = \frac{1}{2}$, find the coordinates of N. Show (in this case) that T and N lie on the line $y = x$ and are such that the product of their distances from the origin is constant.

1.32 Differentiation: Tangents and Normals

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Solutions to Extension Question 1

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